Photonic resistive switching: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01

With $\beta = (2\pi/\lambda_0)(n + i\kappa)$, the field magnitude acquires $e^{-2\pi\kappa L/\lambda_0}$. Squaring gives $T = e^{-4\pi\kappa L/\lambda_0}$. At $\kappa = 0.01$ the exponent is -0.1621, so $T_H \approx 0.8503$; at 0.08 it is -1.2972, so $T_L \approx 0.2733$. Thus $ER \approx 10\log_{10}(0.8503/0.2733) = 4.929\,\mathrm{dB}$ and $IL_H \approx 0.704\,\mathrm{dB}$. The effective mode includes field distribution in multiple materials and boundaries; a bulk index alone does not determine it.

E02

For constant V and I, $E = VI\Delta t = (1.2)(10\times10^{-6})(20\times10^{-9}) = 2.4\times10^{-13}\,\mathrm{J} = 240\,\mathrm{fJ}$. Driver and interconnect loss, optical probe energy, reset and sense circuitry are excluded. A changing waveform requires integrating the measured product instead.

E03

$G = 0.3(77.5\,\mu\mathrm{S}) = 23.25\,\mu\mathrm{S}$, so $R = 1/G \approx 43.0\,\mathrm{k\Omega}$. The pair $T_1 = 0.1, T_2 = 0.2$ gives the same sum and hence the same conductance in the model. A conductance value or step alone does not distinguish these channels or determine a unique atomic arrangement.

E04

The probability of zero failures is $(1-p)^{100}$. Solving $(1-p)^{100} = 0.05$ yields the one-sided 95% upper bound $p = 1 - 0.05^{1/100} \approx 0.0295$, or 2.95%. This calculation assumes independent observations with a common p, and applies to the stated test only. Correlated failures across cycles or devices violate the independence assumption; 100 correlated outcomes cannot simply be inserted in this formula. The result does not prove zero failure or retention at untested times.

Lin Cing https://github.com/xinhi6716/resistive-memory-self-study

Photonic resistive switching: Worked solutions (continued)

E05

Optical transmission depends on mode overlap, absorption, scattering and resonance. A conductive change may add absorption in one waveguide but shift or enable a transmitting resonance in a different MIM structure. Control at least wavelength, input polarization, input power, coupling geometry, temperature and the electrical state preparation before comparing. LRS refers to electrical resistance, not a universal optical ordering.

E06

The absolute difference is $0.25 - 0.20 = 0.05$, i.e. five percentage points. The ratio is $0.25/0.20 = 1.25$, and $10\log_{10}(1.25) \approx 0.969\,\mathrm{dB}$. Without the baseline R_1, the same difference could give a different ratio, so dB cannot be inferred from the difference alone.

E07

A hybrid guide uses a low-index gap between a high-index guiding region and metal; MIM uses an insulating gap between metal boundaries. Both can place optical field near a small active region, but their field distributions and metal overlap depend on their geometry and cannot be ranked without a specified design. Greater field in lossy metal can increase interaction but also dissipate more power. The simple planar-interface TM surface-mode dispersion assumes homogeneous infinite half-spaces and does not describe every finite, layered or patterned guide.

E08

Apply a specified electrical write pulse; record a low-disturbance optical read; set device bias to zero and wait a stated time at controlled temperature; repeat the read and compare with a control and the initial state. Repeat across cycles, devices and wait times while monitoring probe-induced changes. A sweep hysteresis is measured while voltage varies and may include a transient response; it does not establish state survival at zero bias.

Photonic resistive switching: Worked solutions (continued)

E09

From E01, $T_H \approx 0.8503$. Under the stated independent fixed-coupling model, $T_{\text{total}} = T_H^3 \approx 0.615$ and $IL_{\text{total}} = 3(0.704) \approx 2.112\,\text{dB}$. Extra coupler loss, reflections or coherent interference make the simple product insufficient.

E10

$P_{\text{avg}}/f = 12.5 \times 10^{-9}/10^6 = 12.5 \times 10^{-15}\,\text{J} = 12.5\,\text{fJ}$ per operation on average under the stated complete-period measurement and average one-operation-per-period rate. Instantaneous power can vary: its time average already includes active and idle intervals. Peak or active-only power omits part of that averaging and cannot simply replace P_{avg}. This system average need not equal an individual write pulse's device energy. Measure synchronized device voltage and current and integrate $\int V I\, dt$ over the pulse, separately reporting driver and peripheral energy.

E11

The absolute reflectance difference is $0.35 - 0.30 = 0.05$, i.e. **five percentage points**. The corresponding power ratio is $0.35/0.30 = 7/6$, giving $10\log_{10}(7/6) \approx 0.669\,\text{dB}$. These reflectances are hypothetical: a five-point change alone does not determine the ratio without the initial reflectance.

TEM contrast is consistent with local structural or compositional variation; an EDX composition difference provides an additional chemical clue. Together they can support an oxygen-redistribution hypothesis, but neither directly measures vacancy charge states or proves a continuous conductive path. Thickness, preparation artefacts and other local chemical changes must also be considered. For example, compare written and unwritten regions using matched preparation, together with independently calibrated local chemical mapping and spatially registered electrical measurements. Such controls constrain alternatives; they do not automatically establish a unique mechanism.

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